Problems Output Debug Console **Terminal** Ports 2

```
liu@DESKTOP-6872PCL:~/starter$ cmake --preset debug
Preset CMake variables:

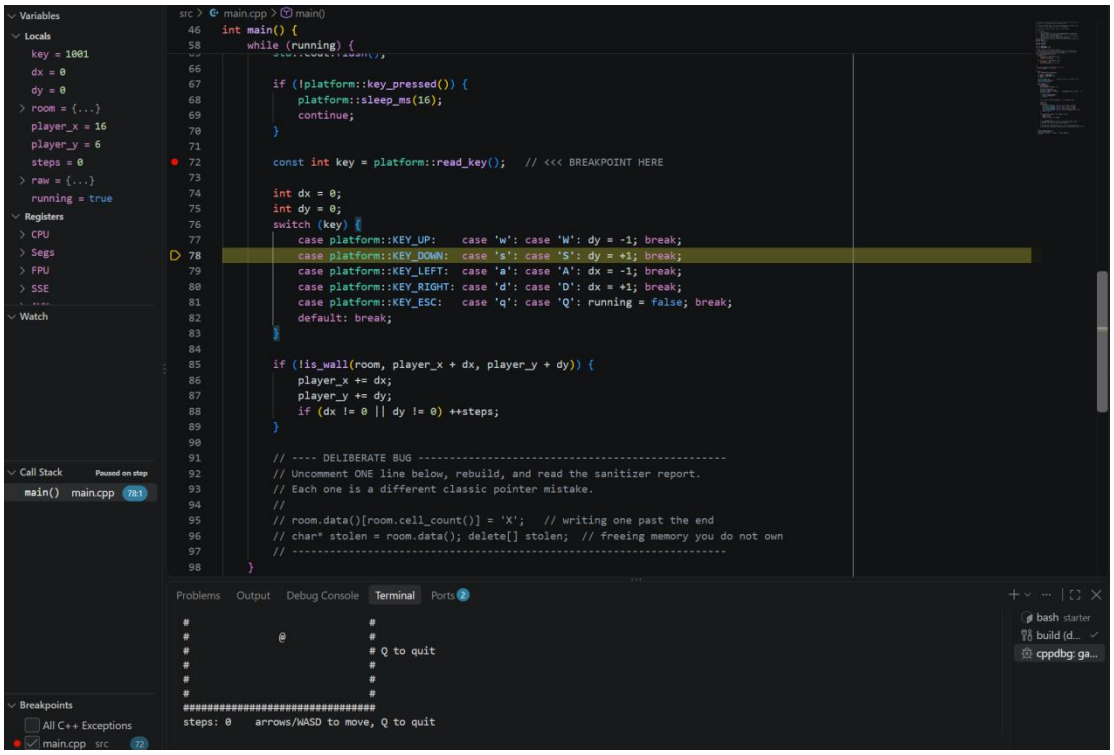
  CMAKE_BUILD_TYPE="Debug"
  COURSE_ENABLE_SANITIZERS="ON"

-- The CXX compiler identification is GNU 14.2.0
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (1.8s)
-- Generating done (0.0s)
-- Build files have been written to: /home/liu/starter/build/debug
liu@DESKTOP-6872PCL:~/starter$ cmake --build --preset debug
[ 14%] Building CXX object CMakeFiles/course_core.dir/src/platform.cpp.o
[ 28%] Building CXX object CMakeFiles/course_core.dir/src/grid.cpp.o
[ 42%] Linking CXX static library libcourse_core.a
[ 42%] Built target course_core
[ 57%] Building CXX object CMakeFiles/game.dir/src/main.cpp.o
[ 71%] Linking CXX executable bin/game
[ 71%] Built target game
[ 85%] Building CXX object tests/CMakeFiles/test_grid.dir/test_grid.cpp.o
[100%] Linking CXX executable ../bin/test_grid
[100%] Built target test_grid
```

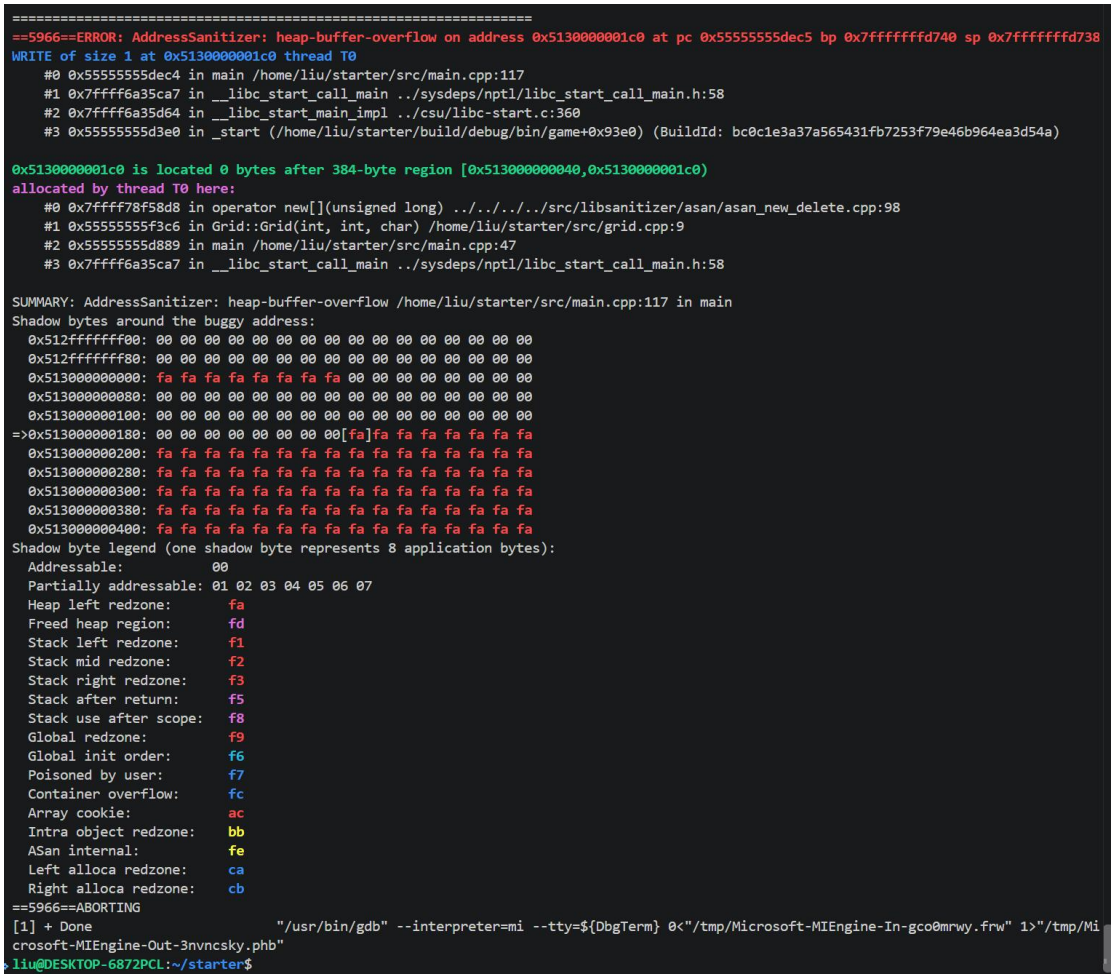
```
liu@DESKTOP-6872PCL:~/starter$ ctest --preset debug
Test project /home/liu/starter/build/debug
    Start 1: test_grid
1/1 Test #1: test_grid ........................   Passed    0.03 sec

100% tests passed, 0 tests failed out of 1

Total Test time (real) =   0.03 sec
```



Answer1:有 0,1000,1001,1002,1003,1004 因为在 platform.h 中解析成了自定义的虚拟码，同时还会等于对应字母的 ASCII 码值。



Answer2:这次错误归咎于 room.date()[room.cell_count()] = 'X';这一行。在这一行中的写入超出了数组末尾下标，最大的下标应该是 cell_count()-1，而这个是合法下标的下一个。